FLEET TEAM DIGEST

737MAX-FTD-31-21002

Issue Title : Nuisance DPC-1 or DPC-2 Status Messages During Power Transfer

Airplane Model	ATA	Minor Model(s)
737MAX	3165-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
04/27/2021	08/27/2026	06/25/2026

Estimated Completion Date	Next Update Date
12/22/2027	12/23/2026

Revision Description

There is no update to this issue at this time.

Applicability

737MAX Airplanes

Description

The Display Processing Computer (DPC) runs a power up Built In Test Equipment (BITE) self-test sequence when it is turned on. This BITE self-test sequence takes about 30 seconds to complete and requires that both the primary and backup power are connected in order to complete the test. This self-test is performed every time the airplane is powered on to ensure the integrity of the DPC.

In the event that the power up BITE self-test is not completed, a non-dispatchable status message 'DPC-1' or 'DPC-2' will annunciate accompanied by the maintenance light. In the event of a 'DPC-1' or 'DPC-2' status message annunciation, one or more of the following Maintenance Messages (MM) may be logged and viewable in fault history:

- 31-02921 DPC-1 INCOMPLETE BITE
- 31-02922 DPC-2 INCOMPLETE BITE
- 31-00241 DPC-1 HAS AN INTERNAL FAULT
- 31-00242 DPC-2 HAS AN INTERNAL FAULT
- 31-90011 DPC-1 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-90012 DPC-2 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-00291 DPC-1 HAS NO INPUT FROM DPC-2 ON CHANNEL 2 X-TALK BUS
- 31-00292 DPC-2 HAS NO INPUT FROM DPC-1 ON CHANNEL 2 X-TALK BUS
- 31-00371 DPC-1 HAS NO INPUT FROM DPC-2 ON FIBER X-TALK CHANNEL A BUS
- 31-00372 DPC-2 HAS NO INPUT FROM DPC-1 ON FIBER X-TALK CHANNEL A BUS

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Boeing has determined this can also happen when the airplane is not connected to external power and an Auxiliary Power Unit (APU) start is initiated from battery prior to the DPC completely coming online to drive the displays. The APU start can draw down the battery voltage enough for the DPC to sense that it has 'lost' one of its sources of power during its power up BITE sequence. This power-on scenario would cause the noted nuisance faults. Note: the faults are not set if the airplane is powered up from external power.

Boeing and Collins Aerospace believe this is a nuisance issue. The DPCs are still operational and as long as the faults clear, the DPCs do not need to be removed.

Status

Boeing has been monitoring Airplane Health Management (AHM) and flight delays reported to Boeing for 'DPC-1' or 'DPC-2' status messages and correlated maintenance messages occasionally during airplane power-on. Boeing is conducting a root cause investigation with the supplier, Collins Aerospace.

Boeing analyzed different sets of DPC shop fault data associated with this nuisance issue. Boeing's analysis of DPC shop fault data indicates that during the power up procedure there is a voltage drop that does not allow the DPC to complete internal power on BITE sequence on start up. This would cause the DPC to annunciate and log a 'DPC-1' or 'DPC-2' status message with any of the correlated fault maintenance messages listed in the description.

At this time, Boeing suspects that the DPC cannot complete internal BITE checks due to either of following.

1. Low battery voltage during power transition.
2. Or power transition operational sequence.

Boeing has published a revision to the Fault Isolation Manual (FIM) for the Maintenance Messages listed below:

- 31-02921 DPC-1 INCOMPLETE BITE
- 31-02922 DPC-2 INCOMPLETE BITE
- 31-00241 DPC-1 HAS AN INTERNAL FAULT
- 31-00242 DPC-2 HAS AN INTERNAL FAULT
- 31-90011 DPC-1 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-90012 DPC-2 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-00291 DPC-1 HAS NO INPUT FROM DPC-2 ON CHANNEL 2 X-TALK BUS
- 31-00292 DPC-2 HAS NO INPUT FROM DPC-1 ON CHANNEL 2 X-TALK BUS
- 31-00371 DPC-1 HAS NO INPUT FROM DPC-2 ON FIBER X-TALK CHANNEL A BUS
- 31-00372 DPC-2 HAS NO INPUT FROM DPC-1 ON FIBER X-TALK CHANNEL A BUS

The FIM revisions have been published and are available in the June 2021 toolbox release (DR21.06).

Boeing published a revision to the Flight Crew Operations Manual (FCOM) to update the Electrical Power Up Supplementary Procedure in the November 2021 FCOM release.

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Boeing initiated a Service Related Probem (SRP) investigation "Avionics Power Transfer and Sequencing Faults Investigation" to support root cause investigation for these nuisance faults.

Interim Action

PREVENTATIVE ACTIONS:

In order to avoid experiencing these nuisance faults, Boeing suggests to perform the following preventative actions:

- Allow the DPC to complete its power on process and drive the displays prior to performing an APU start. This takes approximately 30 seconds once the battery is turned on. See AMM TASK 49-11-00-860-801 for a description of APU Starting and Operation.
- Please note that SAFO 20015, noted in AD 2020-24-02, mandates that the revised aircraft depreservation tasks are adhered to in order to ensure the safe return to service. Upon return to service, complete AMM SUBTASK 10-11-07-960-003 (Active Storage) or AMM SUBTASK 10-12-00-960-002 (Prolonged Storage) to install a new or restored/repaired main battery.
- Ensure the main battery is operational and charged. See AMM TASK 24-31-21-710-801 for a description of Main Battery Charger Operational Test.
- Ensure the Standby Power System is operational. See AMM TASK 24-34-00-710-801 for a description of Standby Power System Operational Test.

Additionally, operators are advised to be aware of the available Collins Aerospace SB DPC-2120-31-1. Per the Collins Aerospace SB DPC-2120-31-1, the expected benefit of the modification is a reduced number of DPC-1 status messages.

For units in warranty, Rockwell Collins will install this service bulletin at no charge for the materials and labor. For units out of warranty, cost of installation is the responsibility of the equipment owner.
NOTE: The in-warranty expiration date for this service bulletin is **May 1, 2023**.

In the event of experiencing these nuisance faults in service, Boeing suggests to cycle the power to the affected DPC by performing the following interim actions at a convenient on-ground maintenance period in the order specified. The following interim actions will be incorporated in the FIM revision targeted for a September 2021 Toolbox release.

INTERIM ACTIONS:

If **DPC-1/-2** is affected, perform the following procedure:

1. Open the primary circuit breaker for DPC-1 C04023 [P18-2, D5] / DPC-2 C04025 [P6-1, D11].
   See AMM SUBTASK 31-65-02-040-001.
2. **Wait at least 20 seconds** for the primary circuits to power down, then open the holdup circuit breaker for DPC-1 C04022 [P6-1, D10] / DPC-2 C04024 [P6-1, D9].

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See AMM SUBTASK 31-65-02-040-002.

3. **Wait at least 20 seconds** for the holdup circuits to power down, then close DPC-1 / DPC-2's holdup circuit breaker, followed by the primary circuit breaker.
   See AMM SUBTASK 31-65-02-860-010.
4. **Wait at least three minutes** for the MAX Display System (MDS) to complete its internal BITE after you apply power. Push the MFD SYS button on the P9-82 Multi-Function Panel (MFP), then push the SEL button on the SELECTOR knob. Make sure that there are no DPC faults shown on the MDS STATUS page.
   See AMM SUBTASK 31-65-02-440-001.

If the nuisance faults clear, the affected DPC does not need to be removed and is operational.

If the faults do not clear, follow AMM procedures for removal and replacement of the affected DPC.

- See AMM TASK 31-65-02-000-801 for a description of Display Processing Computer Removal.
- See AMM TASK 31-65-02-400-801 for a description of Display Processing Computer Installation.

Final Action

Boeing will continue to monitor the 737 MAX fleet for this nuisance condition. Final action is to be determined.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Fault Isolation Manual Revision: June 2021 (Complete)

Flight Crew Operation Manual Revision: November 2021 (Complete)

Root Cause Established: TBD

Solution Selected: TBD

Change Committed: TBD

Production Incorporation: TBD

Service Bulletin Available: TBD

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Operator Action

In the event of experiencing these nuisance faults in service, Boeing requests operators to submit the following information to Boeing via Boeing Communication System (BCS) Service Request (SR) under ATA 31.

- Boeing requests operators to please provide the datum aircraft and flight information during which the nuisance fault occurred.
- Boeing requests operators please provide a DPC shop fault download from both DPCs on the datum aircraft per AMM TASK 31-65-02-470-802. This data may be submitted via Message Courier, addressed to "VM-787 ATA 45/46 Service Engineering". Please include the reference SR message number (i.e. XXX-XXX-12-3456) in the message title.
- Boeing requests operators to please provide any pilot write-up, flight logbook entries, or maintenance write-ups related to what was observed when the nuisance faults took place.

Affected Documents

Boeing published FIM revisions for the following MMs in the June 2021 Toolbox release of the FIM:

- 31-02922 DPC-2 INCOMPLETE BITE
- 31-00241 DPC-1 HAS AN INTERNAL FAULT
- 31-00242 DPC-2 HAS AN INTERNAL FAULT
- 31-90011 DPC-1 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-90012 DPC-2 HAS NO OUTPUT ON MORE THAN ONE BUS
- 31-00291 DPC-1 HAS NO INPUT FROM DPC-2 ON CHANNEL 2 X-TALK BUS
- 31-00292 DPC-2 HAS NO INPUT FROM DPC-1 ON CHANNEL 2 X-TALK BUS
- 31-00371 DPC-1 HAS NO INPUT FROM DPC-2 ON FIBER X-TALK CHANNEL A BUS
- 31-00372 DPC-2 HAS NO INPUT FROM DPC-1 ON FIBER X-TALK CHANNEL A BUS

Boeing published a FCOM revision updating the Electrical Power Up Supplementary Procedure in the November 2021 FCOM release.

Supplier Information

Rockwell Collins

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TELEPHONE: 1.888.265.5467

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Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737MAX-SRP-27-0325		
Service Bulletin (SB)	DPC-2120-31-1		08/20/2018

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

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